

Updates to CDS and NCIS

By Paediatrics SAG

Slides prepared by Dr Poon Sher Lynn

7th Oct 2020

Changes to DA Schedule

(Age Range)	Recommended Touchpoints by Doctor	
4 – 8 weeks	• 4 weeks DA	same
	• 8 weeks (DA if not previously done)	
3 – 5 months	• 4 months DA	Previously 3 months DA
6 – 12 months	• 6 months - Screen by nurse, refer to Dr if deemed necessary	Previously 9 months DA by nurse
	• 12 months DA	NEW
15 – 22 months	• 18 months DA	same
24 – 36 months	• 30 months DA	NEW , previously 3 year old DA by nurse
4-5 years	• 4 years DA	same

Changes to DA Schedule

- No more 9 month old hearing screening by nurses
 - CDS and NCIS subsidies are now extended to participating CHAS GPs and they do not have similar tools
 - Any suspicions of hearing loss can be picked up through the current developmental screening questions eg.
 - Not startling or turning to sounds
 - Speech delays

Changes to IMM Schedule

ANNEX A

National Childhood Immunisation Schedule (NCIS) (from birth to age 17 years, effective from 1 November 2020)

Vaccine	Birth	2 months	4 Months	6 months	12 months	15 months	18 months	2-4 years	5-9 years	10-11 years	12-13 years	13-14 years	15-17 years
Bacillus Calmette-Guérin (BCG)	D1												
Hepatitis B (HepB)	D1	D2		D3									
Diphtheria, tetanus and acellular pertussis (paediatric) (DTaP)		D1	D2	D3			B1						
Tetanus, reduced diphtheria and acellular pertussis (Tdap)										B2			
Inactivated poliovirus (IPV)		D1	D2	D3			B1			B2			
Haemophilus influenzae type b (Hib)		D1	D2	D3			B1						
Pneumococcal conjugate (PCV10 or PCV13)			D1	D2	B1								
Pneumococcal polysaccharide (PPSV23)								One or two doses for children and adolescents age 2-17 years with specific medical condition or indication.					
Measles, mumps and rubella (MMR)					D1	D2							
Varicella (VAR)					D1	D2							
Human papillomavirus (HPV2 or HPV4)											D1 (Females)	D2 (Females)	
Influenza (INF)		Annual vaccination or per season for <u>all children</u> age 6 months to <5 years (6-59 months).							Annual vaccination or per season for children and adolescents age 5-17 years with specific medical condition or indication.				



Recommended ages and doses for all children



Recommended for persons with specific medical condition or indication

FOOTNOTES:

- **D1, D2, D3:** Dose 1, dose 2, dose 3
- **B1, B2:** Booster 1, booster 2
- **10-11, 12-13, 13-14 years:** Primary 5, Secondary 1, Secondary 2 (Tdap, IPV, HPV (for females) and MMR (as catch-up) vaccines are provided as part of Health Promotion Board's school-based vaccination programme)
- **HepB:** Doses 2 and 3 are recommended to be given as part of the 6-in-1 vaccine at 2 and 6 months, respectively
- **MMR:** Only the dose 2 is recommended to be given as part of the MMRV vaccine

Summary of Changes to NCIS

- **6-in-1** (inclusive of Hep B) to replace 5-in-1 vaccine at 2-months and 6-months.
 - Note: *Infant born to Hep B +ve mother: Hep B (D2) will remain at one month old*
- Addition of **Varicella** vaccines
 - Separate MMR vaccine and varicella vaccine at 12-months-old
 - Combined MMRV at 15-months-old
- **PCV** will sync with 6-in-1
 - Dose 1 at **4-months**, Dose 2 at **6-months**.
- **Influenza** (annual or per season) is now incorporated as part of NCIS from 6 months to 59 months.

NCIS and CDS changes from 1 Nov 2020

Age	NCIS Vaccination	Recommended CDS touchpoints	Age range for CDS touchpoint	NUP Schedule	NUP Schedule for children born to Hep B carrier mother
At Birth*	BCG (D1) Hep B (D1)	-	-		
4 weeks	-	4 weeks	[1] 4 – 8 weeks	Dr DA	Hep B (D2) + Dr DA
2 months	6-in-1 (D1) †	-		6-in-1 (D1) (optional Rotarix D1) DA (only if not done at 4 weeks)	5-in1 (D1) (optional Rotarix D1) DA (only if not done at 4 weeks)
3 months	-	-	[2] 3 – 5 months	-	-
4 months	5-in-1 (D2) PCV (D1)	3 months or 4 months**		5-in-1 (D2) PCV (D1) (optional Rotarix D2) Dr DA	
5 months	-	-		-	
6 months	6-in-1 (D3) PCV (D2)	6 months	[3 & 4] 6 – 12 months	6-in-1 (D3) PCV (D2)	
12 months	MMR (D1) Varicella (D1) PCV (B1)	12 months		MMR (D1) Varicella (D1) PCV (B1) Dr DA	
15 months	MMRV (D2)	-	[5] 15 – 22 months	MMRV (D2) (offer INF D1)	
18 months	5-in-1 (B1)	18 months		5-in-1 (B1) (INF D2) Dr DA	
30 months	-	30 months	[6] 24 – 36 months	30 months DA (offer INF) Nur DA + Dr DA	
48 months	-	48 months	[7] 48 – 60 months	48 months DA (offer INF) Nur DA + Dr DA	

Plan FLU (D1) to offer at 15 months and FLU (D2) at 18 months (If given for the first time at 15 months). Subsequent, annual flu jab only need one dose

Why keep the first visit at 4 weeks?

- It is important to pick up abnormalities that may not show up or be obvious at birth eg.
 - Cardiac murmurs
 - Congenital cataracts
- To pick up prolonged jaundice early and refer appropriately
- Monitor weight
- Feeding issues

Transition

- From August 2020, Nurses have started booking patients on the new schedule
- Also offering short deferments in immunisations appropriately for parents who prefer to enjoy the new subsidies

Transition

- Note: Children who were last seen at 18 months old in July 2020 and scheduled for next **DA at 3 years old (36 months old)** up to Jan 2022 will not be rescheduled → convert to a Drs DA when they come
- Children seen at 18mo in Aug 2020 booked to **new DA schedule at 30 months old** (Aug 2021)
- There will be an overlap period where we will be assessing some children at 30 months and some at 36 months, need to adjust DA expectations appropriately.

EPIC DA Flowsheet

ake/Output IV Assessment Daily Cares/Safety Vitals-OP DA - NNJ to < 3m DA - 3 to 12 months DA - 15mths to 6 years

Accordion Expanded View All 1m 5m 10m 15m 30m 1h 2h 4h 8h 24h Based On: 07:0

View All

Plan

Doctor's Assessment

Developmental assessment done at

Concerns from developmental checklist in health booklet?

Other significant history

Physical Examination

Growth Parameters (Height/Weight/OFC)	?	Abnormal
Head - Fontanelles	?	Abnormal
Eyes - Cornea/Light reflex/Red Reflex/Squint/Eye Movements		
Ears response		
Mouth/Teeth		
Heart/Lungs		Normal
Femoral pulses	?	Abnormal
Abdomen/Umbilicus		Normal
Back/Spine/Limbs		
Hips		Normal
Genitalia		Female - normal
Skin		
Gait		
Others/Remarks		Line 1 line 2 lin...

Patient Outreac...
22/07/20
14:00

EPIC DA Flowsheet

- Allows ease of continuity between each DA touchpoint
- Allows ease of communication between Nurses and Drs
 - Nurse inputs for same touchpoint easily seen on one flowsheet
- Abnormal values flagged up in red
- Comments box to elaborate on findings (has no formatting)

DA Recap

Corrected Age

- Age of baby adjusted for prematurity.
 - Use EDD or gestational age of 40 weeks as reference.

Corrected age = Chronological age - [40 weeks - gestational age at birth]

Table 2: Recommended ages for CDS, growth monitoring and vaccination based on gestational period

Gestation period	CDS	Growth monitoring	Vaccination
Born before 35 weeks	Corrected age till 24 months old	Corrected age till 24 months old	Chronological age
Born between 35 and 37 weeks	Chronological age	Corrected age till 24 months old	Chronological age
Born after 37 weeks	Chronological age	Chronological age	Chronological age

- Plot corrected growth monitoring chart in EPIC if needed

At All Touchpoints

- Growth Monitoring
- Review red flags listed as “parental concerns” in the HB
- Ensure Developmental Checklists completed
- Ensure Child Safety Checklists completed
- Address parental concerns
- Physical Examination
- Fill and sign the Health Booklet

4 Weeks DA

- Do a **full detailed DA and physical examination** (even if this is not the first visit)
- In particular, look out for
 - Growth – ensure good weight gain
 - Congenital cataracts - ensure bilateral red reflex present
 - Prolonged jaundice
 - Cardiac murmurs
 - Hip dysplasia
 - Feeding issues – breast feeding/vomiting/reflux

4 Weeks DA – First visit history

- Review Antenatal, Birth history

- Gestation
- Birth weight
- Birth history
- TSH level
- G6PD status
- Newborn hearing screening
- Phototherapy done?
- Latest SB level

BIRTH RECORD AND PARTICULARS OF CHILD

Name of child (in BLOCK LETTERS): _____

Birth Certificate No.:

Date of Birth: Time of Birth: hrs

Address: _____

Place of Delivery: _____

Sex: ☐ Male ☐ Female Ethnic Group: _____

Duration of Gestation: Weeks

Mode of Delivery: ☐ Normal ☐ LSCS ☐ Vacuum extraction ☐ Forceps ☐ Other

Apgar Score: 1 min 5 min

Weight at Birth: gm

Length at Birth: cm

Head Circumference: cm

PARTICULARS OF PARENTS

MOTHER

Name: _____ NRIC/Passport No.: _____

Occupation: _____

Tel (RES): _____ Tel (OFF): _____ Tel (HP): _____

FATHER

Name: _____ NRIC/Passport No.: _____

Occupation: _____

Tel (RES): _____ Tel (OFF): _____ Tel (HP): _____

SIGNIFICANT EVENTS DURING PREGNANCY / DELIVERY

Jaundice ☐ No ☐ Yes Phototherapy ☐ Yes Exchange Transfusion ☐ Yes

NEWBORN SCREENING

G6PD Deficiency ☐ No ☐ Yes

TSH: _____ mIU/L TT4: _____ pmol/L Date: _____

TSH Screening Done ☐ No ☐ Yes Date: _____

Hearing Screening

☐ ** OAE Date: _____ ☐ ** ABAER Date: _____

Left Pass: ☐ No ☐ Yes Left Pass: ☐ No ☐ Yes

Right Pass: ☐ No ☐ Yes Right Pass: ☐ No ☐ Yes

Needs further evaluation: ☐ No ☐ Yes

Remarks (if any): _____

INVESTIGATION(S) DONE (if any)

Serum Bilirubin (highest level): _____ μ mol/L Date: _____

Blood Group: _____ Date: _____

Other Tests (please specify): _____ Date: _____

INFORMATION ON DISCHARGE FROM HOSPITAL

Date: _____ Weight: gm Breast Feeding: ☐ Yes ☐ No

Serum Bilirubin (if done) before discharge: _____ μ mol/L

Instructions to doctors and nurses:
All weight, length and head circumference measurements are to be entered on the charts on pages 26-41.
Please document additional medical findings in the summary of clinic/hospital medical record section on pages 10-13.
*The newborn/Early of Neonatal, **OAE: Otoacoustic Emission, and ***ABAER: Automated Brainstem Auditory Evoked Response.

- Any specialist follow up? → pg 60 of HB

4 Weeks DA – Normal Development

- Babies may be older eg. 8 weeks old by the time they come for immunization appt. Tailor your DA accordingly.

(Age Range)	Recommended touchpoints	Milestones achieved by 90%	
4 – 8 weeks	4 weeks	Gross Motor	Equal movements Lifts head momentarily when prone
		Fine Motor	Grasp Reflex
		Speech	Alerts to sound
		Social	Regards faces Spontaneous social smile Barely follows moving object
	8 weeks	Follows object to mid-line Vocalises	

4 Months DA – Normal Development

(Age Range)	Recommended touchpoint	Milestones achieved by 90%	
3 – 5 months	4 months	Gross Motor	Prone head up 45° Props up on forearms No head lag
		Fine Motor	Unfists mostly Hand regard/inspects fingers/hands to mouth Grasps objects in hand
		Speech	Vocalises when talked to
		Social	Turns to sound out of sight Follows object past midline

6 months old

- Screening questions by nurses

6 months	1. Growth monitoring : weight, length, OFC* 2. Feeding history 3. Parents/Caregivers please answer the questions below***: • Can your child roll over? Yes/No • Can your child turn towards a sound? Yes/No • Can your child reach out for things? Yes/No
----------	--

12 Months DA – Normal Development

(Age Range)	Recommended touchpoint	Milestones achieved by 90%	
6-12 months	12 months	Gross Motor	Crawls Sits unsupported Stands with support Pulls to stand
		Fine Motor	Transfer objects Finger thumb grasp Bangs 2 cubes held in hands
		Speech	Turns to sound that is out of sight Single syllables “ma”, “da”, “ba”, “ga” Imitates speech sounds
		Social	Looks for fallen object Reacts to stranger Claps hands

18 Months DA – Normal Development

(Age Range)	Recommended touchpoint	Milestones achieved by 90%	
15 – 22 months	18 months	Gross Motor	Stoops to recover Walks well independently
		Fine Motor	Pincer Grasp Scribbles Builds a tower of 2 cubes
		Speech	Says papa/mama specifically
		Social	Imitates household activities Indicates wants by gestures and pointing Understands simple instructions

30 Months DA– Normal Development

(Age Range)	Recommended touchpoints	Milestones achieved by 90%	
24 – 36 months	30 months	Speech	Points to 2-4 pictures correctly Names 2 pictures correctly Two word phrases
		Social	Removes clothing Imaginative play eg. Comb doll's hair
		Fine Motor	Builds tower of 4-6 cubes
		Gross Motor	Walks up and down steps Kicks ball forward

4 Years DA – Normal Development

(Age Range)	Recommended touchpoints	Milestones achieved by 90%	
48 – 60 months	4 years	Gross Motor	Pedals tricycle
		Fine Motor	Draws straight line Copies a circle
		Speech	Knows name/age/sex Names a friend Speaks 4-5 word, full sentences
		Social	Bladder control by day Plays with friends Follows 3 step instructions

Where and When to Refer

- Summarised from MOH Circular

Refer to General Paediatrics

- **Gross motor delay with hypertonia/hypotonia**
- **Differences between right and left side of body in strength, movement, tone or function**
- **Regression of motor milestones at any age (even if within normal range)**
- **Regression of milestones in multiple domains**
- **Isolated abnormal OFC (esp. if disproportionate to weight/height)**
- **Children with abnormal growth and clinical concerns of medical/syndromic/genetic condition**
 - Otherwise, advise appropriately and monitor for 6 months before referring

Where and When to Refer

Refer to Child Development Unit

- **Language delay**
- **Language regression**
- **Autism red flags**
- Other developmental concerns
- **Delay** in multiple domains

• Lack of response to sound • Lack of response to visual stimuli	Refer Paeds ENT Refer Paeds Eye
• Universal Newborn Hearing Screening not done or abnormal	Refer Paeds ENT or call hospital's UNHS programme for appt (KKH: 63941893; SGH: 63214540; NUH: 97248736)

Future Developments

- NUP DA Guide to be released when approved
- Standardisation of red flags by MOH-to be released when ready
- Revision of 2014 Health Booklet by Health Promotion Board
- MCHAT R/F study 18th month- ongoing

Q&A

Thank You



National University
Hospital

Developmental Assessment and Red Flags

Dr. Tan Mae Yue
Associate Consultant
Child Development Unit
KTP-NUCMI, National University Hospital Singapore
4 November 2020



<restricted copy>



National University
Hospital

Outline

- Childhood development: Importance
- Developmental assessment
- And associated red flags
- Regression
- Case studies:
- What to do after the red flags?



MINISTRY OF HEALTH
SINGAPORE

MH 69:01

MOH Circular No. 183/2020

13 July 2020

All Registered Medical Practitioners

GUIDANCE ON CHILDHOOD DEVELOPMENTAL SCREENING

Early childhood programmes are important to ensure that children receive the best possible start and to facilitate early detection and timely intervention.

<restricted copy>

Child Development

Continuous but predictably **sequential** biological, psychological and emotional changes that occur in human beings between birth and the end of adolescence.

The sequence of development is the same for all children and can be described in terms of developmental milestones.

As children develop at different rates, which are determined by a complex interplay of environmental and genetic factors, the age of attainment for each milestone ranges widely.

Poll 1

How common is Developmental Delay?

- 1-5%
- 10-15%
- 25-30%
- 40%
- 50%

Developmental delays are common in childhood, occurring in 10%–15% of preschool children. Global developmental delays are less common, occurring in 1%–3% of preschool children.

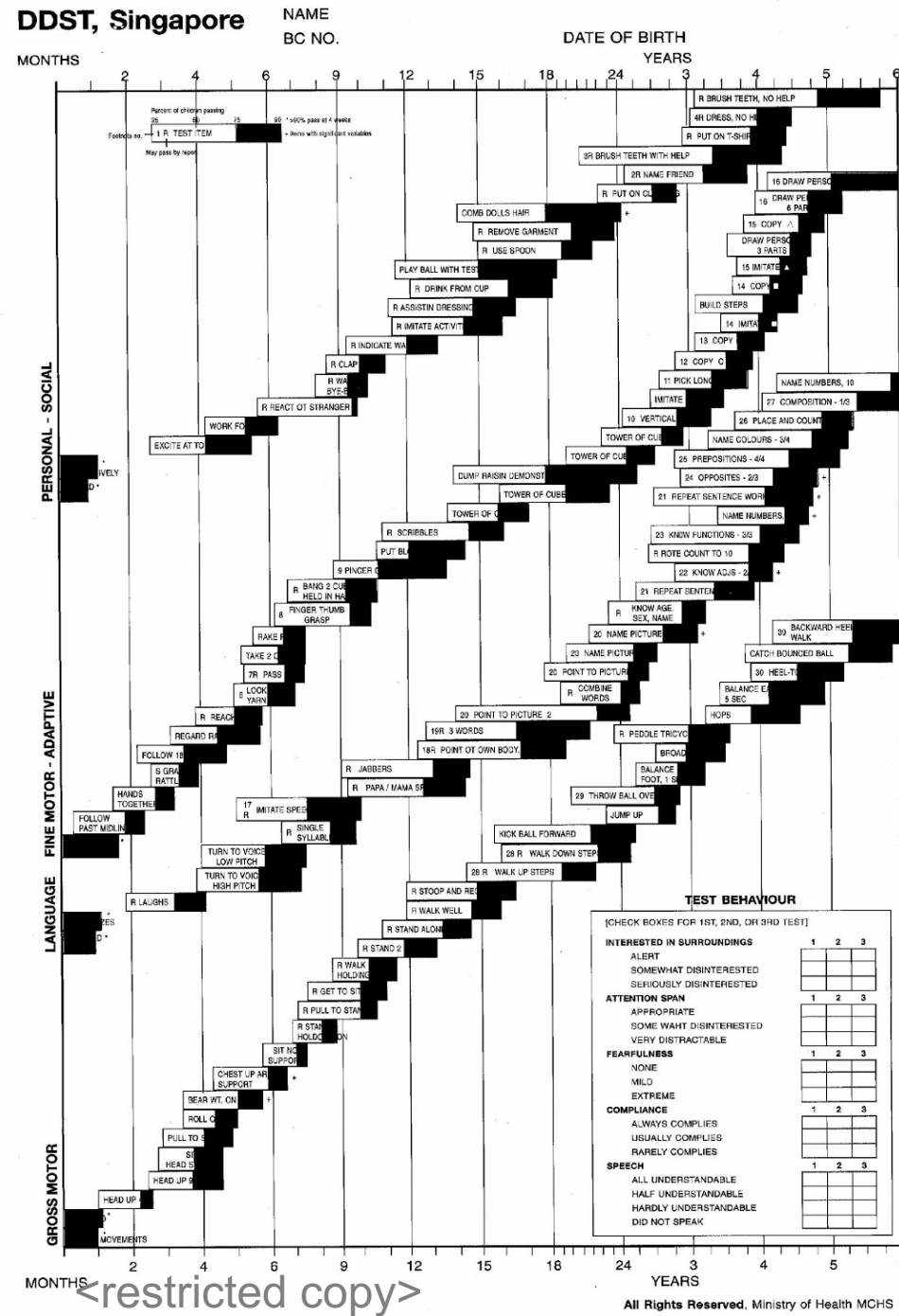
Childhood developmental screening aims to identify concerns or delays in development during the early years.

[The Importance of Early Intervention for Infants and Toddlers with Disabilities and Their Families pdf icon\[1.19 MB, 2 Pages\]external icon](#) Accessed February 26, 2015.

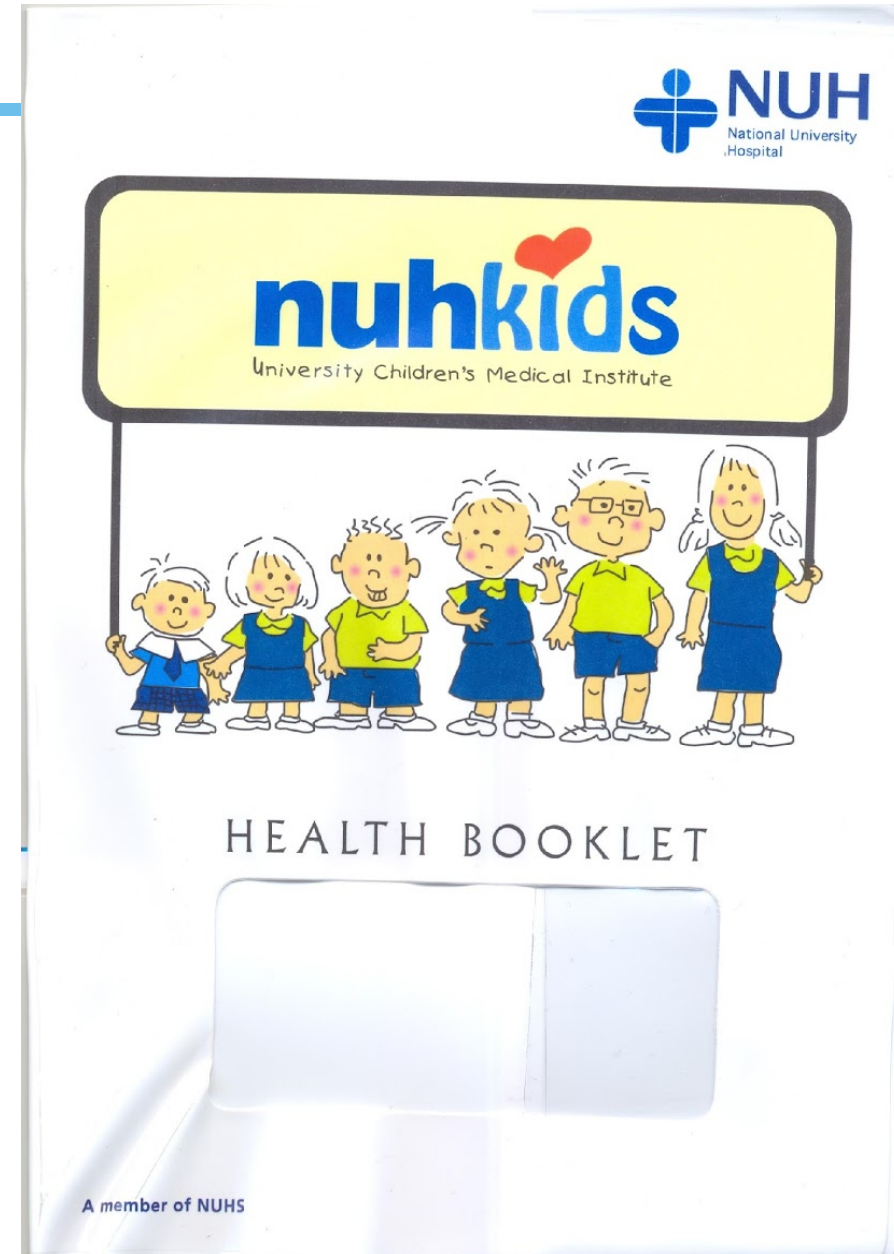
Why is detecting developmental delay important?

Investing during early childhood development is the **most cost effective period in the overall management of disability**

The Denver Development Screen Singapore



Singapore Child Health Booklet



<restricted copy>

Poll 2

How many recommended touchpoints are there for well child checks at the polyclinic?

- 4
- 5
- 6
- 7
- 8

Touchpoints – primary care

	Recommended touchpoints	Age range
1	4 weeks	4-8 weeks
2	3-4 months	3-5 months
3	6 months	6-12 months
4	12 months	6-12 months
5	18 months	15-22 months
6	30 months	24- 36 months
7	48 months	48-60 months

The domains

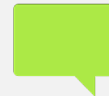
**RED FLAG
WARNING**



Gross motor



Fine motor / Vision



Speech and Language / Hearing



Social skills

Developmental Assessment

<restricted copy>

Approach to Developmental Delay

1. Developmental History

- ***Eliciting concerns from parents and carers**
 - *Sensitivity and specificity as good as standardised screening instruments (Glascoe et al. Clin Paeds 2003, 42: 133-138)*
 - ***Knowing normal developmental milestones**
 - Health booklet
 - Various instruments
 - www.cdc.gov/actearly
 - ***Recognising red flags**
- Examination by Structured Observation

- Define what is the delay- what tools to use?
- How bad is the delay ? Age equivalent
- Vision
- Hearing
- (what is the cause)
- (other domains)

General Examination

- Growth centiles:
 - height, weight, head circumference
- Dysmorphic features
- Neurological examination

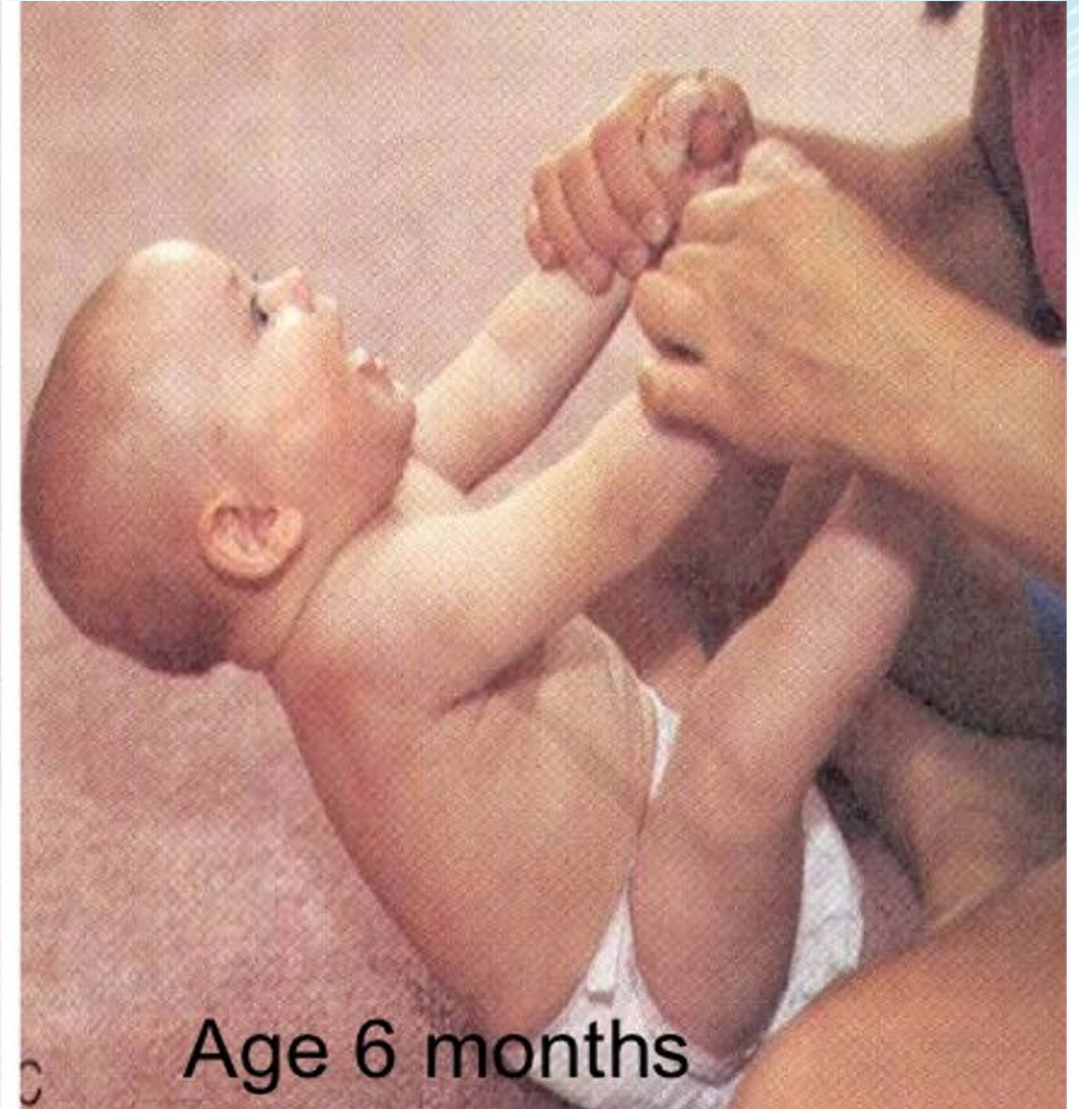
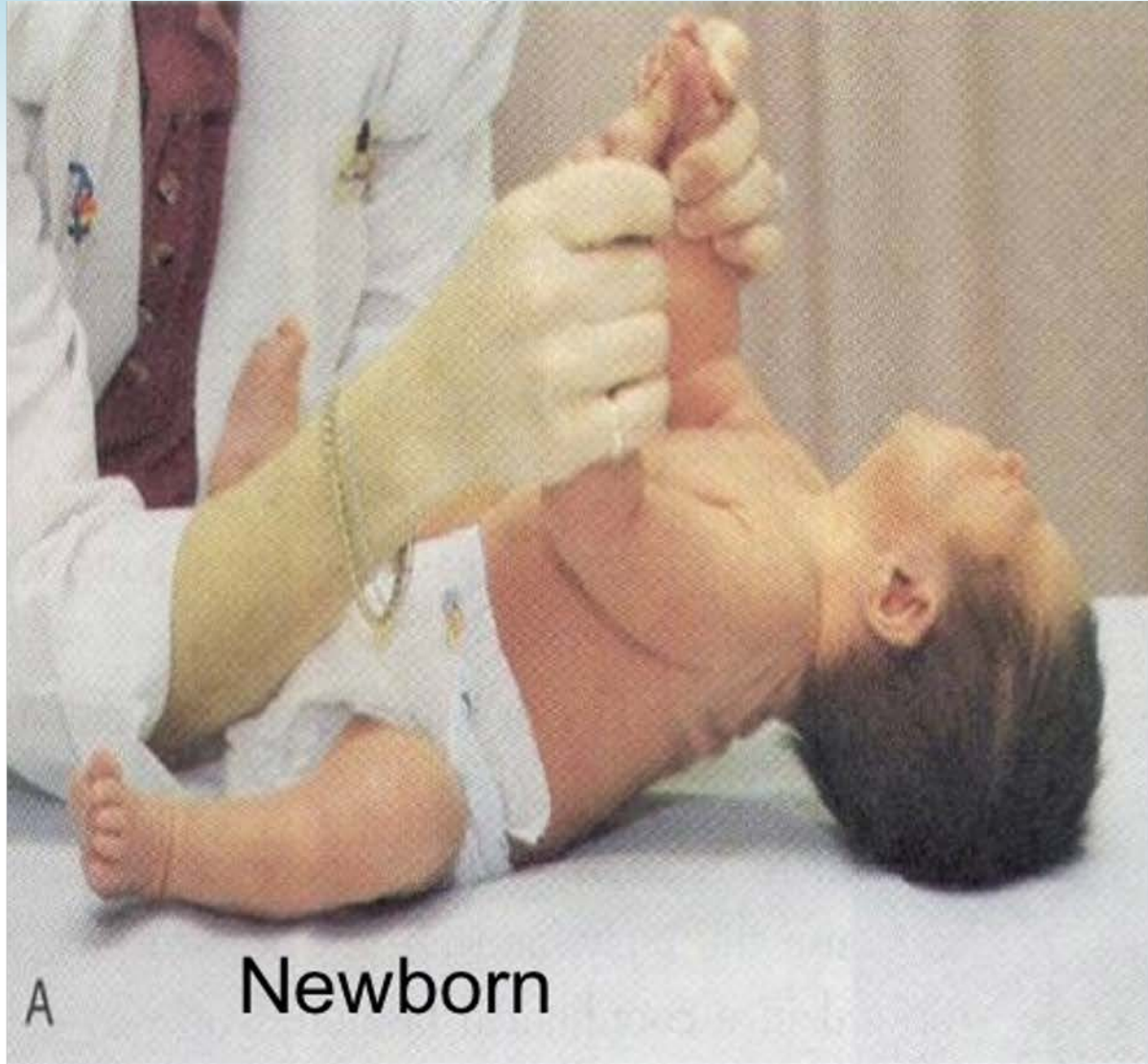
Gross motor

- Acquisition of gross motor skills **precedes** the development of fine motor skills
- Both occur in cephalocaudal fashion
 - head control preceding arm and head control
 - followed by leg and foot control

Gross motor development

- Newborn: barely able to lift head
- 6 months: easily lifts head, chest and upper abdomen and can bear weight on arms

Head control



<restricted copy>

Gross motor development

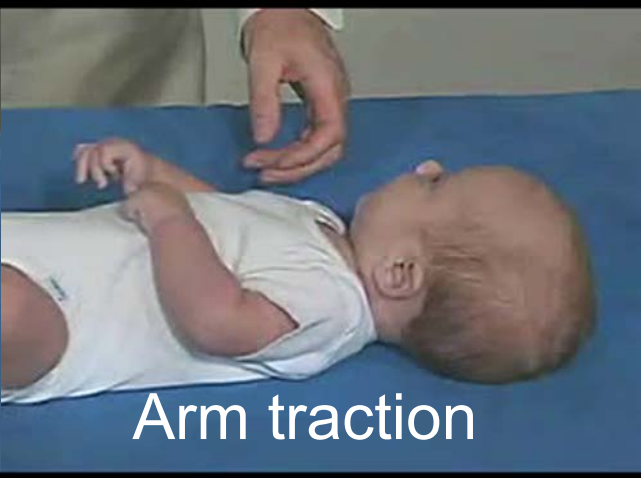
- 2 month: Head up when prone
- 3 month: Head control
- 5 to 6 month: Rolls over, sits tripod
- 9 month: Sits well independently

Commando crawl

Pull to stand



Supine



Arm traction

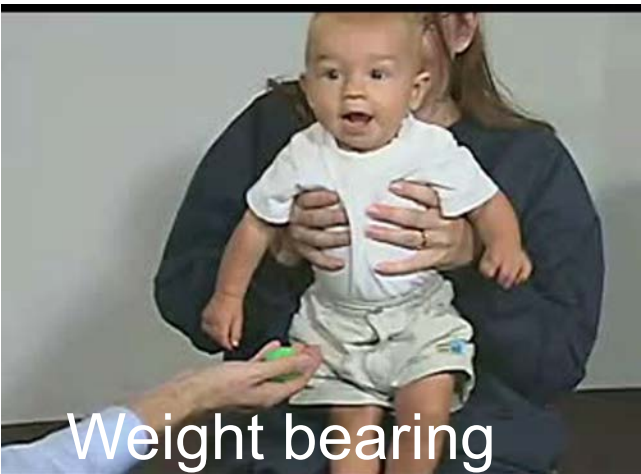


Sitting



Prone

The 180° examination



Weight bearing



Ventral suspension



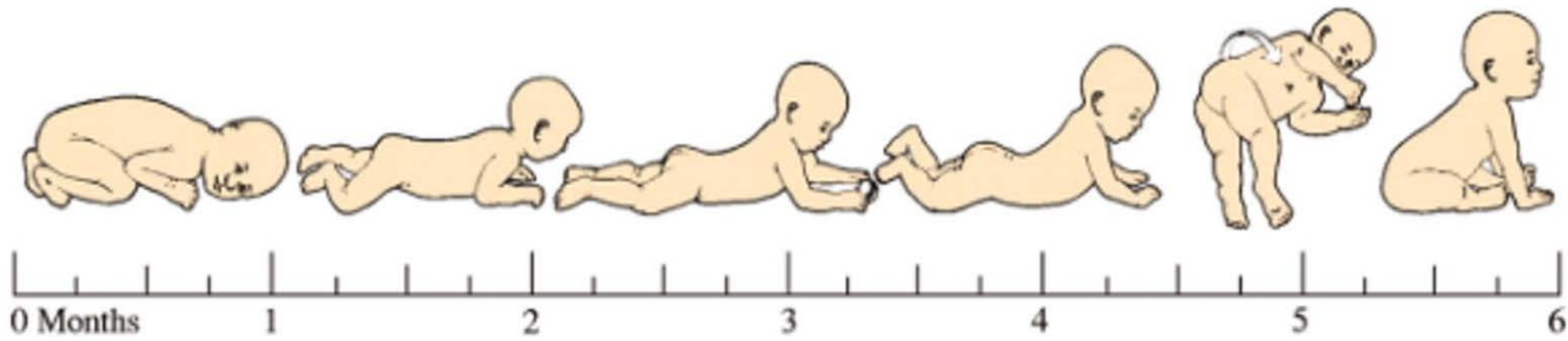
Parachute

Gross motor development

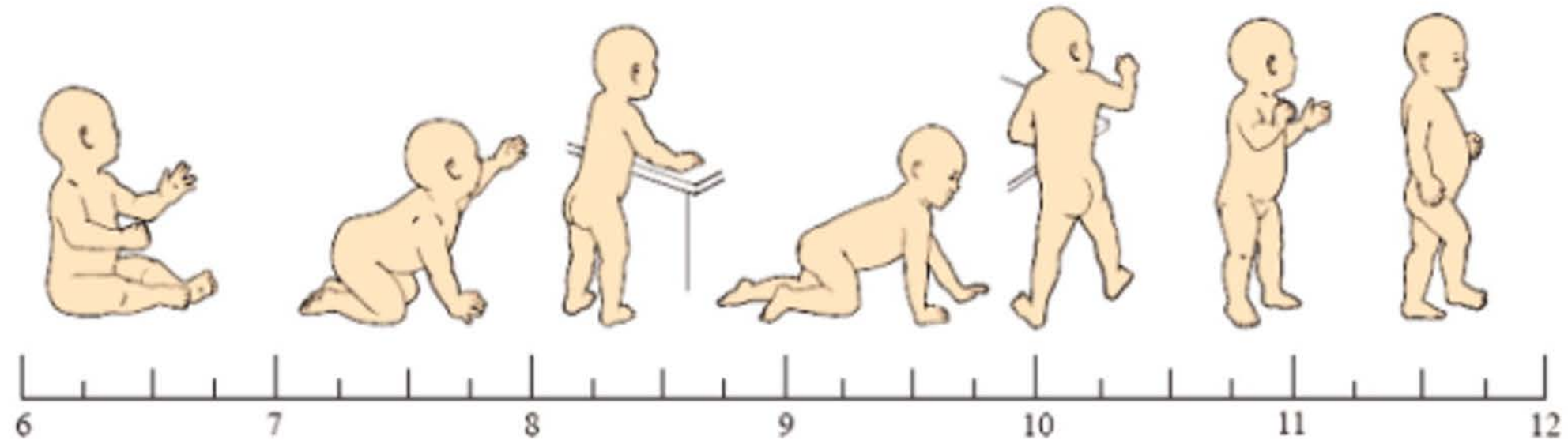
- 10 to 12 month: Stand without support,
Cruising
- 13 to 14 month: Walks independently (most)
- 18 month: Walks up stairs with 1 hand held

Gross motor development

- 2 years: Walks up and down stairs with both feet on same step, kicks ball, throws ball overhead, walks backwards
- 2.5 years: Walks up and down stairs with alternate feet, runs well
- 3 years: Jumps, rides tricycle
- 4 years: Stand on 1 foot, throws, catches and bounces ball
- 5 years: Hops on 1 foot at least 5 steps, skips



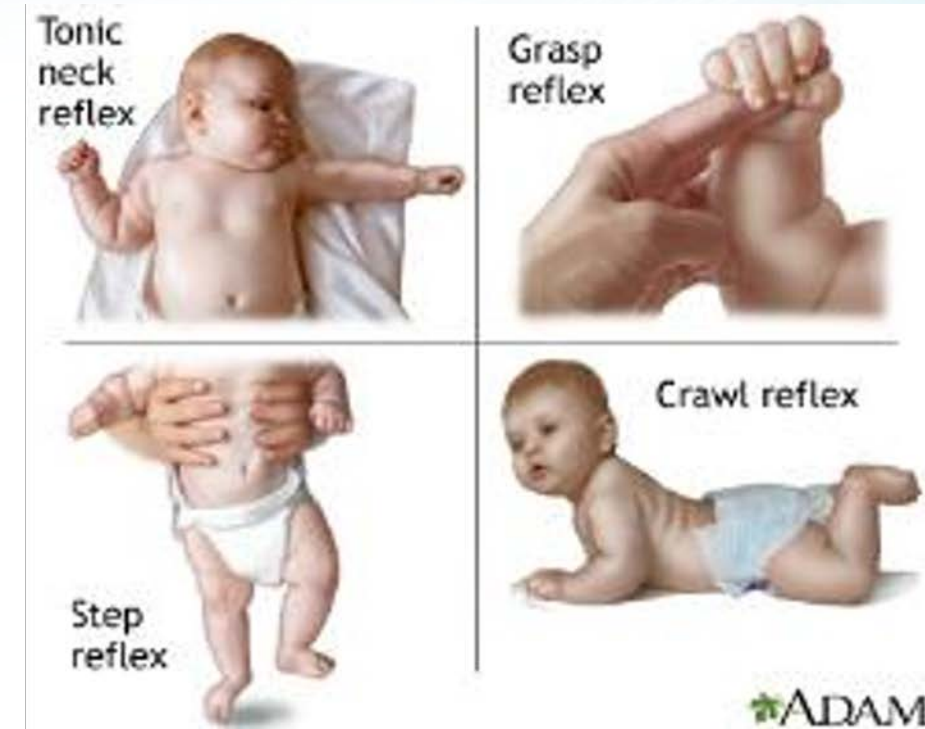
Chronologic progression of gross motor development



<restricted copy>

Primitive reflexes

1. Moro reflex
 - starts in utero, inhibited by 16 weeks after birth
2. Palmar grasp
 - starts in utero, inhibited 2-3 months after birth
3. Asymmetric Tonic Neck Reflex
 - starts in utero, disappears by 6 months after birth
4. Stepping reflex
 - appears at birth, lasts for 3-4 months, then reappears at 12-24 months
5. Landau's reflex (back arches and head rises when prone)- starts 3 months of life, lasts until 12m



Poll 3

A 6 month old child comes for a routine OPS review. Which of the following is a red flag?

1. Unable to roll over
2. Unable to sit
3. Has an upgoing plantar reflex
4. Has an ATNR

Red Flags: Gross Motor

- Floppy baby
- Head lag ... 3 months
- poor head control at ... 4 months
- Unable to roll over, persistence of primitive reflexes beyond... 6 months
- persistence of flexor hypertonia in lower limbs beyond... 9 mths
- not sitting independently with straight spine by.. 10 months
- not walking by ... 18 months
- clumsiness, poor balance, asymmetrical signs

Fine motor and Vision

- 4 to 6 weeks: Hands fisted, visual fixation and tracking
- 3 months: Hand unfisted
- 5 months: Reaches out for objects
- 6 to 7 months: Transfers objects from hand to hand



<restricted copy>

Fine motor and vision

- 9 months: Immature pincer approach
- 10 to 12 months: Mature pincer grasp, turns pages of a book
- 15 months: Build tower of two blocks
- 18 months: Build tower of three blocks, scribbles, develops hand preference



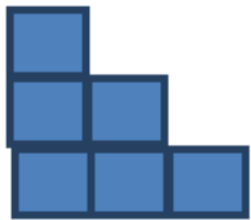
<restricted copy>

Know your tools!

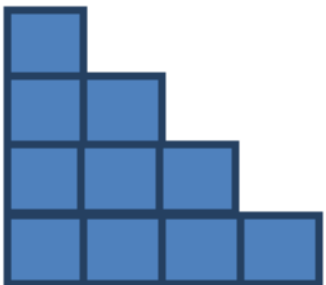


<restricted copy>

Blocks



4y



5y



15m



18m



2y



2y



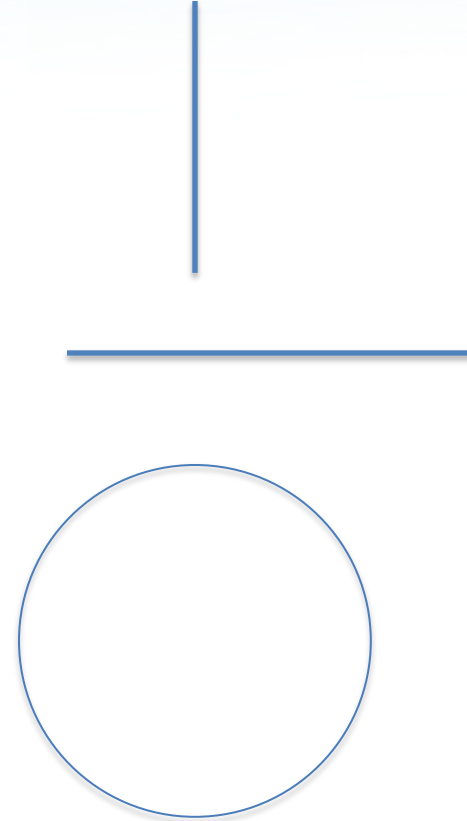
3y



- Bang two cubes: 1yo
- Stacks 2 cubes: 15mo
- Stacks 3 cubes: 18mo
- Stacks 6 cubes or train: 2yo
- Stacks 8 cubes or train with chimney: 2.5yo
- Stacks 9 cubes or bridge (with hole): 3yo
- Able to build a 3-step wall with 6 cubes: 4yo
- Able to build a 4-step wall with 10 cubes: 5yo

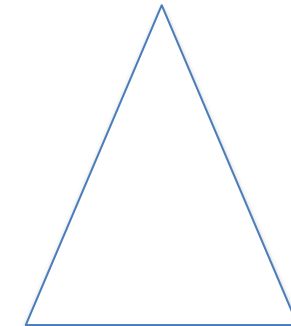
Fine motor and vision

- 2 years: Copies straight line
Immature grasp of pen
Snipping
- 2.5 years: Copies horizontal line
- 3 years: Copies circle
Later copies cross
Beading/threading
Cut straight line



Fine motor and vision

- 4 years: Copies square
Draws stick figure of man with head, trunk and body parts
- 5 years: Copies triangle
Draws figure of man with facial features
(no fingers or toes)



Poll 4

Before what age is hand preference a concern?

- A. 12 months
- B. 15 months
- C. 18 months
- D. 24 months

Red Flags: Fine Motor + vision

- Does not fix and follow **3 months**
- not reaching with hands at **6 months**
- persistence of hand regard after **9 months**
- no index finger exploration/ exploration of toys by **12 mths**
- early hand preference before **12 mths**
- no scribbling by **2 years**
- Does not copy line by **3 years**, circle by **4 years**
- no tripod grasp of pencil by **4 years**

TYPICAL PENCIL GRASP DEVELOPMENT FOR HANDWRITING

10 Months
Pincer Grasp



12-15 Months
Palmar Supinate Grasp



2-3 Years
Digital Pronate Grasp



3-4 Years
Quadrupod Grasp



3-4 Years
Static Tripod Grasp



5-6 Years
Dynamic Tripod Grasp



Speech and Language and Hearing

- 4 to 6 weeks: Smiles in response to voice
- 3 months: Coos and laughs
- 5 to 6 months: Localises sound on the level of the ear, then above ear
- 9 months: Babbles, mimics sounds



Speech and Language and Hearing

- 10 to 12 months: First words, understands “no”
- 12 months: Looks at familiar family member when names, ‘mama’, ‘papa’, has 2-3 words
- 15 months: Obeys 1 step commands
- 18 months: Points to 3 body parts, points to self, 10 to 25 word vocabulary

Speech and Language and Hearing

- 2 years: 2-3 word phrases, 50+ words, can point to >5 body parts,
- 2.5 years: knows name
- 3 years: knows name, age and gender, speaks in sentences, asks many questions
- 4 years: tells a story, sings nursery rhymes
- 5 years: names 4-5 colours, counts to 20

Social

- 4 to 6 weeks: social smile
- 3 months: gazes at mother when feeding
- 5 to 6 months: puts things in mouth
- 7 months: stranger anxiety
- 9 months: object permanence, plays peek-a-boo

Social

- 10 to 12 months: claps, waves bye-bye, feeds self finger food
- 15 months: drinks from cup
- 18 months: takes off socks or shoes, some pretend play, able to point and gesture

Social

- 2 years: start to indicate toilet needs
- 3 years: some parallel games with other children, help in dressing, can do buttons, dry by day and night, feed self with utensils
- 4 years: plays with social interaction and role playing, can take turns, dresses self well, brushes teeth
- 5 years: bathes and showers without help

Poll 5

Which of the following is a red flag?

- A. Cooing only at 3 months
- B. Babbling only at 12 months
- C. 'Papa', 'Mama' only at 14 months
- D. Knows name and age only at 3 years old

Red Flags: Speech and Language

- no smile/ vocalising by **3 mths**
- no babbling by **10 mths**
- not responding to name by **12 mths**
- no clear spontaneous words by **18 months**
- no understanding of common objects by **18 months** eg. give me the cup
- no word combinations by **30 mths**
- no routine use of 2-3 word sentences by **3 yr old**
- no understanding of what/why questions at **4 yr old**
- **echolalia**

Red Flags: Social and Play

- no chuckling by **5mths**
- no interest in social games by **10mths**
- lack of interest in toys by **18 months**
- absence of pretend play by **24 months**
- preference for solitary play by **3-4 yrs old**
- **no eye contact, repetitive play**
- **limited imaginative play, not joining others in imaginary play**

*2 Important Questions to ask

1. Is the developmental delay restricted to **specific domains or affects more than 1 domains (global)**?
1. Is development **delayed** or **regressing**?

CASE STUDIES

Case 1: 6 months review

- Child came for routine check and vaccine at 6 months
- Nurse refers this patient to you from DA room- as child still unable to roll or sit with arm support.

Child doesn't like tummy time since young. Unable to roll over.

Fine motor- child about to grab objects, palmar grasp

Speech: babbling, can smile with parents

Born full term, well after birth

No family history of neurodevelopmental disorders

Physical examination

Frog legged posture

Head lag

Slip sign

Sits with rounded back, parent holding

What else would you examine for?

Video showing tongue fasciculations

What would you do? Poll 6

- A. TCU next routine check
- B. TCU in 1 month to monitor gross motor milestones
- C. Refer to paediatrics
- D. Refer to CDU

Case 2: K

Now 18 months girl

Child doesn't want to stand anymore

- Could walk at 15 months
- Became 'weak' and now cannot sit unsupported
- Cannot stand or walk
- Used to be able to throw things, now can lift hands but no more pincer grip

Was able to babble → now just social smile

Further history

- Abnormal eye movements
- Visual fixation poor
- Stiffening movements when crying

Further history

- 1st child
- Mother 40 years old
- Father 52 years old
- FT LSCS for prev myomectomy
- BW 3.6kg, APGAR 9,9
- Birth TSH, G6PD normal, not sure about IEM
- No issues in infancy, and well until 2 months ago

Physical examination

- Not dysmorphic, OFC 47cm 25th centile, same as weight/height
- No NC stigmata
- Fixate and track but dysconjugate gaze, nystagmus – multidirectional
- Low tone, floppy, slip sign+
- Hyperreflexia, upgoing plantars
- H/L NAD, No organomegaly

Poll 6

Which of the following is a red flag?

- A. Floppy
- B. Regression
- C. Nystagmus
- D. All of the above

**What is your
initial
impression?**

Global developmental delay

+

- Neurological symptoms
- Macrocephaly/microcephaly,
- Hepatosplenomegaly
- Dysmorphisms
- Café-au-lait/neurocutaneous

GDD with neuroregression

GDD Assessment

With neuroregression

**+ neuro signs: ? Seizure? Opsoclonus? Hypotonia
and hyperreflexia**

Case 3: 18 month check

Child now needing support to walk in last 2-3 weeks

Wobbly walking still

Previously could walk independently well

Now will ask to be pulled up to walk

18 month check: history

- Term infant, well at birth, BW 3060
- 2 older brothers, well
- No trauma
- Gradual history of not wanting to walk, wobbly
- Could run unsteadily before this well

Video showing unsteady trunk

Child falling down when trying to fall

No nystagmus

Poll 7: What is your assessment?

- a. Delayed gross motor milestone
- b. Gross motor regression
- c. Cerebellar signs

Progress

Impression in OPS

New onset LL weakness TRO Neuromuscular disorder

Early appointment

Child seen in 1 week

- Frank trunk titubation then
- Eventual MRI brain> posterior fossa tumour

Case 4: 1y 7 month check

Mom feels she has no major concerns but agrees he is different from twin brother

Teachers had asked mom to get referral.

Child loves running, runs around in class on his own

Further review of milestones: 24 month check

- GM: run, started skate scooter
- FM: scribbling, mom not sure if can draw lines
- Receptive speech: 1 step command
- Expressive: 2 word phrases

Poll 8

What would you do next?

- A. Discuss speech stimulation
- B. Ask the mother to enroll the child in school now
- C. Refer to Paediatrics
- D. Refer to CDU

Where and when to refer

Refer to General Paediatrics

- GM delay with hyper/hypotonia
- Asymmetry in right/left side of body in strength/movement/function/tone
- Regression of motor milestones at any age
- Regression in multiple domains
- Isolated abnormal OFC (esp if disproportionate)
- Children with abnormal growth AND clinical concerns of medical/syndrome/genetic conditions

Where and when to refer

Refer to CDU

- Language delay
- Language regression
- Autism red flags
- Other developmental concerns
- Delay in multiple domains

Other

- Lack of response to sound > Paeds ENT
- Lack of response to visual stimuli > Paeds Eye
- Newborn hearing not done > Paeds ENT, or call hospitals UNHS

<restricted copy>

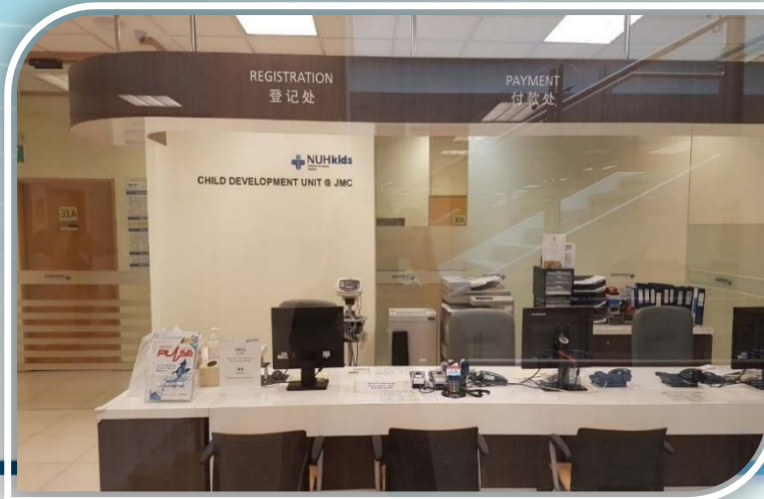
How to refer a child to CDU?

Subsidised Rates	Private Rates
Eligible for Singaporeans and Permanent Residents when referred by: • Any polyclinic • Any restructured hospital (e.g. NUH) where you are receiving treatment at subsidised rates • A General Practitioner under the Community Health Assist Scheme (CHAS)	Non-subsidised rates apply if you: • Call us directly for a self-referral • Referred by a General Practitioner or private hospital • Referred from a restructured hospital where you are a private patient

Note: Consultations at CDU are by appointment only.

Child Development Unit

For children < 7 years old



CDU @ Jurong Medical Centre
60 Jurong West Central 3, Level 2,
Singapore 648346
☎: +(65) 6665 2530/2531
Email: cdu@nuhs.edu.sg

CDU @ Keat Hong
2 Choa Chu Kang Loop, #03-01,
Singapore 689687
☎: +(65) 6769 4537/4637
Email: cdu@nuhs.edu.sg



<restricted copy>